ECONOMETRICA
JOURNAL OF THE ECONOMETRIC SOCIETY

When are Variance Ratio Tests for Serial Dependence Optimal?

Author(s): Jon Faust

Source: *Econometrica*, Sep., 1992, Vol. 60, No. 5 (Sep., 1992), pp. 1215-1226

Published by: The Econometric Society

Stable URL: https://www.jstor.org/stable/2951545

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

*The Econometric Society* is collaborating with JSTOR to digitize, preserve and extend access to *Econometrica*

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

*Econometrica*, Vol. 60, No. 5 (September, 1992), 1215–1226

# WHEN ARE VARIANCE RATIO TESTS FOR SERIAL DEPENDENCE OPTIMAL?

BY JON FAUST[1]

This paper considers a class of statistics that can be written as the ratio of the sample variance of a filtered time series to the sample variance of the original series. Any such statistic is shown to be optimal under normality for testing a null of white noise against some class of serially dependent alternatives. A simple characterization of the class of alternative models is provided in terms of the filter upon which the statistic is based. These results are applied to demonstrate that a variance ratio test for mean reversion is an optimal test for mean reversion and to illustrate the forms of mean reversion it is best at detecting.

KEYWORDS: Variance ratio statistic, serial dependence, optimal test.

## 1. INTRODUCTION

MANY TEST STATISTICS for time series dependence can be interpreted as variance ratios, and it has long been known that such tests can be optimal tests of certain hypotheses. Durbin and Watson (1950), for example, took T. W. Anderson's (1948) general results on this topic as a starting point in deriving their test for first-order dependence. However, many of these theoretical results on the optimality of variance ratios provide no simple answers to two practical questions. First, given a variance ratio statistic, how does one determine the null and alternative hypotheses that the statistic is best for testing? Second, given a null and alternative, what variance ratio test should one use?

The inability to answer questions such as these has complicated the implementation and interpretation of a popular variance ratio test for mean reversion (see, e.g., Cochrane (1988), Cogley (1988), Kim, et. al. (1988), Lo and MacKinlay (1988, 1989), Poterba and Summers (1988), Richardson and Smith (1989)). Uncertainty about the test's optimality in detecting mean reversion has led to questions about the importance of test results. Further, implementation of the test requires selecting a maximum order of serial correlation to consider, and it has been unclear how the optimality properties of the test vary with this choice.

This paper sets out and analyzes a class of variance ratio statistics for which many of these issues are easily resolved. Statistics in this class are the ratio of the sample variance of a filtered time series to the sample variance of the original series and will be called *filter variance ratio* (FVR) statistics. The class includes the variance ratio test for mean reversion and the Durbin-Watson statistic. Two central results are provided. First, each FVR test is shown to share the large sample properties of a likelihood ratio test for a certain null and alternative. Second, a simple relation between the filter a statistic is based on and the forms of dependence against which the test has optimal power is

[1] Much of this work was completed while the author was at the Federal Reserve Bank of Kansas City and the Economics Department at Princeton. I thank Sean Becketti, Gregory Chow, Tim Cogley, Neil Ericsson, and anonymous referees for valuable comments.



This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

presented. This relation is used to establish heretofore unrecognized optimality properties of the variance ratio test for mean reversion.

## 2. FVR TESTS AND THEIR PROPERTIES

To any finite-order moving average filter, $\phi(L)$, there corresponds an FVR statistic that is of the form

$$\operatorname{var}(R^{\phi})/\operatorname{var}(R)$$

where $\operatorname{var}(\cdot)$ stands for an estimate of variance, $R$ is a $(T \times 1)$ time series, and $R^{\phi}$ is a $m$th-order moving average filter of $R$ with elements

$$R_t^{\phi} = \sum_{i=0}^{m} \phi_i R_{t-i} = \phi(L) R_t.$$

It is useful to write the statistic as a ratio of quadratic forms; to do so, write the vector $R^{\phi}$ as $\Phi R$ where $\Phi$ $((T-m) \times T)$ is given by

$$\Phi = \begin{bmatrix} \phi_m & \cdots & \phi_0 & 0 & \cdots & 0 \\ 0 & \ddots & & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & & \ddots & 0 \\ 0 & \cdots & 0 & \phi_m & \cdots & \phi_0 \end{bmatrix}.$$

Next, in order to simplify the proofs, pick a slight modification of the standard sample variance for the numerator of the statistic. Defining $\bar{R}$ as the sample mean of $R$, estimate the mean of $R^{\phi}$ by $\phi(1)\bar{R}$.[2] This allows the sample variance of $R^{\phi}$ to be written $(R - \bar{R}i)'\Phi'\Phi(R - \bar{R}i)/T$ (where $i$ is a conformable vector of ones), suggesting the variance ratio statistic

$$\nu_{\phi} = \frac{(R - \bar{R}i)'\Phi'\Phi(R - \bar{R}i)}{(R - \bar{R}i)'(R - \bar{R}i)}. \tag{1}$$

The Durbin-Watson (DW) statistic provides the simplest example of an FVR statistic. This statistic is defined as $\Sigma_{t=2}^{T}(\hat{\varepsilon}_t - \hat{\varepsilon}_{t-1})^2/\Sigma_{t=1}^{T}\hat{\varepsilon}_t^2$, where the $\hat{\varepsilon}_t$ are residuals from some regression. If the regression has a constant, then the sample mean of $\hat{\varepsilon}$ is zero, and the DW statistic is an FVR statistic based on the filter $\phi(L) = (1 - 1L)$.

The variance ratio test for mean reversion is also based on an FVR statistic. The statistic is defined as the ratio of the sample variance of an $m+1$ period difference of a time series, $Y$, to $m+1$ times the sample variance of the first difference of $Y$:

$$V(m) = (m+1)^{-1} \frac{\operatorname{var}(Y_t - Y_{t-m-1})}{\operatorname{var}(Y_t - Y_{t-1})}. \tag{2}$$

[2] Thus, $\phi(1)\bar{R}i = \Phi i\bar{R}$. Using this mean instead of the sample mean of $R^{\phi}$ in the numerator simplifies the proofs, but all results are valid with the statistic defined as the ratio of the standard sample variances or with the statistic defined as a linear combination of the standard sample autocorrelations consistent with (1).

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

To see $V(m)$ as $(m+1)^{-1}$ times an FVR statistic, define $R_t = Y_t - Y_{t-1}$, and note that an $m+1$ period difference of $Y$ is an $m$ period moving sum of $R$. Thus, taking the filter, $v(L) = 1 + L + L^2 + \cdots + L^m$,

$$V(m) = (m+1)^{-1}\operatorname{var}(R^v)/\operatorname{var}(R).$$

Cochrane (1988) has noted that the variance ratio test for mean reversion can be written as a weighted sum of the first $m$ autocorrelations. Algebraic manipulation of (1) reveals that this property carries over to all FVR statistics. Thus, the results below for FVR statistics apply to a range of statistics that are weighted sums of autocorrelations, such as many spectral density-based statistics.

### 2.1 *Hypotheses for which FVR Tests Are Optimal*

This section shows that every FVR statistic will have optimal power for testing certain parametric null and alternative hypotheses. The work of Anderson (1948) on quadratic forms and the work of Davidson and MacKinnon (1987) regarding the implicit null and alternative of tests foreshadow this result—the trick is matching the statistic with the null and alternative it is best for testing.

The basic results shown in Propositions 1 and 2 below are these: Each FVR statistic is optimal under normality for testing the null hypothesis that a data series is not serially correlated, against the alternative that it comes from a particular class of autoregressive (AR) models. The class of alternatives that the FVR statistic based on $\phi(L)$ is best at detecting will be called the *$\phi$-best* class. If $\phi(L)$ is of order $m$, each member of the $\phi$-best class is an AR($m$) process characterized by (3) below.

A sketch of the proof proceeds by defining the $\phi$-best class and then deriving the FVR test as an approximate equivalent to the likelihood ratio (LR) test for best-class alternatives. The $\phi$-best class includes those AR($m$) processes with AR polynomials, $\rho_q(L)$, that for some $q$ solve

$$\rho_q(z)\rho_q(z^{-1}) = \alpha\big(1 + q\phi(z)\phi(z^{-1})\big) \tag{3}$$

for $z$ inside the unit circle and some $\alpha > 0$. The process is indexed by $q$ and not $\alpha$ because $\alpha$ merely scales the process (changes the variance). Notice that this class nests white noise as a special case, $q = 0$.

The proofs are in the time domain and rely on the following time domain analog of (3) (justified in Lemma 1 of the Appendix). If $\sigma^2\Sigma_q$ is the covariance matrix for a model satisfying (3), then for some $\alpha$

$$\Sigma_q^{-1} \approx \alpha(I + q\Phi'\Phi). \tag{4}$$

These matrices are equal except for the $(m \times m)$ blocks in the northwest and southeast corners. This equation reveals the fundamental insight that $\Phi'\Phi$, the matrix in the numerator of the FVR statistic, is linearly related to the inverse covariance matrix of a class of AR models. This linearity is crucial below, allowing simple factorization of an approximate LR test.

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

Consider a test of the null that a series is uncorrelated against the alternative that it is generated by some best-class model with $q > 0$. This is a one-sided test of the single restriction $q = 0$ versus $q > 0$ on (3). The goal is to show that the FVR test is an approximate LR test of these hypotheses; the proof proceeds from a test of the point alternative hypothesis, $q = q^* > 0$, to a general result that is independent of the particular $q^*$ chosen. Thus, the hypotheses of interest are

$$H_N\colon R \sim N(\delta, \sigma^2 I),$$

$$H_K\colon R \sim N\left(\delta, \sigma^2 \Sigma_{q^*}\right),$$

where $\Sigma_{q^*}$ is the covariance matrix of the $\phi$-best class process with $q = q^*$. The parameters $\delta$ and $\sigma^2$ are unconstrained by both hypotheses.

The LR principle suggests basing a test on (some monotonic transform of) the ratio of the sample variances under the null and the alternative. Thus, the null is rejected when

$$LR = \frac{(R - i\tilde{R})' \Sigma_{q^*}^{-1} (R - i\tilde{R})}{(R - i\bar{R})'(R - i\bar{R})} < \gamma_1 \tag{5}$$

for some $\gamma_1$, where $\tilde{R}$ is the maximum likelihood estimate of the mean of $R$ under the alternative (the generalized least squares estimate of the mean). Two steps of approximation get us from the LR statistic to the FVR statistic. First, substitute the sample mean for $\tilde{R}$ in the numerator. This substitution will not affect large sample properties since the sample mean is consistent under the alternative. Second, using (4), substitute $\alpha(I + q^* \Phi' \Phi)$ for $\Sigma_{q^*}^{-1}$. This substitution will also be asymptotically unimportant since the two corner blocks by which the matrices differ affect only the first and last $m$ observations in the numerator quadratic form. These substitutions give the approximate LR test,

$$\frac{\alpha(R - \bar{R}i)'(I + q^* \Phi' \Phi)(R - \bar{R}i)}{(R - \bar{R}i)'(R - \bar{R}i)} < \gamma_2.$$

Divide by $\alpha$, subtract 1, and divide by $q^*$ to give the FVR test that rejects when

$$\nu_\phi = \frac{(R - \bar{R}i)' \Phi' \Phi (R - \bar{R}i)}{(R - \bar{R}i)'(R - \bar{R}i)} < \gamma_3. \tag{6}$$

Notice that the FVR statistic does not involve $q^*$. Since this result is independent of $q^*$, this statistic and test are appropriate for the test of the one-sided alternative $q > 0$.[3] Thus, the FVR test should share the large sample properties of the LR test for both one-sided alternatives.

[3] It is worth emphasizing that since the FVR statistic does not involve $q^*$, its distribution under the null is not affected by $q^*$. Since critical values depend only on the distribution under the null, any critical value, $\gamma_3$, will result in the same test size against all alternatives $q > 0$. The argument can be repeated for alternatives $q < 0$—the test simply rejects for large values of the statistic in this case.

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

Propositions 1 and 2 codify this argument. Proposition 1 establishes the link between the filter $\phi(L)$ and the best-class models by showing that (3) does define a nonempty class of AR($m$) models. Proposition 2 confirms that the FVR test does have the large sample properties of the LR test against best-class alternatives. The proofs are in the Appendix.

PROPOSITION 1: *For any filter $\phi(L)$ of order $m$, if $q > -1/\max_\lambda |\phi(e^{-i\lambda})|^2$, there exists a stationary AR($m$) process with AR polynomial $\rho_q(L)$ solving* (3) *for $z$ inside the unit circle and some $\alpha > 0$.*

PROPOSITION 2: *Consider any filter $\phi(L)$ of order $m$ with associated FVR statistic $\nu_\phi$. Assuming normally distributed underlying data, if the null is that the data are iid, then:*

(i) *For every $\phi$-best class alternative, there exist an $\alpha$ and $q$ such that $\sqrt{T}(\alpha(1+q\nu_\phi) - LR) \overset{a.s.}{\to} 0$ under both the null and alternative hypotheses.*

(ii) *The FVR test based on $\nu_\phi$ is locally most powerful invariant against alternatives in the best class of the form $q_T^* = q/\sqrt{T}$.*

(iii) *The FVR test based on $\nu_\phi$ has maximum exact slope against nonlocal alternatives in the $\phi$-best class.*

Overall, FVR statistics converge to LR statistics and share both the local and nonlocal large-sample optimality properties of LR tests. These properties hold against the entire $\phi$-best class. As for finite-sample properties, it is useful to note that the FVR statistic differs from (a monotonic transform of) the LR statistic primarily in its treatment of the first and last $m$ observations in the numerator of the statistic. The effect of treating these observations less than optimally in the FVR statistic should rapidly diminish. If $m/T$ is relatively small, it is likely that FVR tests will nearly share the LR test's finite-sample optimality properties.

The robustness of these optimality results to deviations from the assumption of constant variance and normality is of considerable interest. While there are few analytic results in this regard, the following result, suggested by Richardson and Smith (1989), provides a weak form of robustness:

*A two-tailed FVR test with maximum approximate slope given the iid normal null hypothesis will also have maximum approximate slope given any assumptions that leave the sample autocorrelations asymptotically standard normal under the null of no serial correlation.*[4]

This result allows, for example, deviations from normality, but it does not allow conditional heteroskedasticity (Richardson and Smith (1989)). The pri-

[4] Geweke (1981) showed that the approximate slope of a statistic, $\nu^*$, that is asymptotically $\chi^2_{(1)}$ under the null is plim $\nu^*/T$. Proof of the claim requires showing that a two-tailed test can be based on an FVR statistic that is asymptotically $\chi^2_{(1)}$ under both the more and less restrictive nulls. Since $\nu_\phi$ can be written as a linear combination of autocorrelations, the stated restriction implies that under either null, $T^{1/2}(\nu_\phi - \mu) \overset{a}{\sim} N(0, \sigma_\phi^2)$, where $\mu$ and $\sigma_\phi^2$ depend on the filter coefficients. A two-tailed test can be based on the upper tail of $\nu^* = T(\nu_\phi - \mu)^2/\sigma_\phi^2$, which is asymptotically $\chi^2_{(1)}$ under either null.

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

mary limitation of this result, however, is that the approximate slope is a misleading proxy for the exact slope in many contexts (Bahadur (1967)).

### 2.2 *Critical Values for FVR Tests*

A side benefit of writing FVR statistics as ratios of quadratic forms is that the exact distribution of such statistics under normality is known. The distribution can easily be evaluated using numerical approaches due to Imhof (1961) or Pan Jie-Jian (1968), and reviewed by Durbin and Watson (1971). For sample sizes under 300 or so, $p$-values can be calculated rapidly on a personal computer.

The experience with the FVR test for mean reversion suggests two justifications for examining critical values premised on underlying normality. First, while approximations have been derived that are valid under a large range of distributional assumptions and under weak dependence and heteroskedasticity (Cogley (1988), Faust (1988), Lo and MacKinlay (1989)), the results regarding reliability of these approximations are mixed, especially when $m/T$ exceeds $1/4$ or so. Given this fact, calculating both the asymptotic $p$-value and the exact value under normality is useful, since the exact $p$-value assuming normality reveals the often substantial magnitude of the approximation error in the asymptotic $p$-value if the data were approximately normal.

The second justification comes from Richardson and Stock's (1989) approximation to the distribution of the mean reversion statistic based on allowing $m$ to grow with $T$ ($m/T \to \delta > 0$ as $T \to \infty$). Their results indicate that the nominal $p$-values using this approximation—which are valid in large samples in the presence of deviations from normality and limited conditional heteroskedasticity—are very close to the exact critical values under normality.[5]

## 3. APPLICATION: THE FVR TEST FOR MEAN REVERSION

A great many studies in the last few years have applied FVR tests for mean reversion based on the variance ratio $V(m)$ in (2). Despite the test's widespread use, however, what form of mean reversion, if any, the test is best against has not been identified, and it has been necessary to rely on Monte Carlo experiments to determine critical values, to determine how to select $m$, and to evaluate the test's power.

The FVR framework provides a more satisfactory approach to many of these issues. The definition of the $\phi$-best class in (3) reveals that the FVR test based on $\phi(L)$ will be optimal for testing the white noise null against AR models with

[5] To be more consistent with Richardson and Stock's approach, one might calculate the exact critical value for the largest $T$ one can accommodate (300 or so on a conventional PC), setting $m$ so that $m/T$ is the same as in the application. In Stock and Richardson's work, however, the critical values did not change much once $T$ was above 100. Evaluating the exact distribution in this case represents a huge time advantage over running the sequence of Monte Carlo experiments proposed by Stock and Richardson.

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

spectra proportional to

$$(7) \qquad s(\lambda) \equiv \left[1 + q\phi(e^{-i\lambda})\phi(e^{i\lambda})\right]^{-1}$$

for any $q$ satisfying Proposition 1. The spectrum of any stationary, finite-order AR process can be represented by (7)—this converse of Proposition 1 is noted in the proposition's proof. Given that any stationary, invertible ARMA model can be approximated arbitrarily well by a finite AR model, and that every finite AR model falls in the best class of some FVR statistic, the FVR class includes nearly optimal tests for any form of ARMA dependence. In practice, of course, the order of the filter is limited by sample size, limiting the quality of the approximation to arbitrary ARMA models.

To apply (7) to the FVR test for mean reversion, simply substitute the filter $V(m)$ that is based on, $v(L) = 1 + L + \cdots + L^m$, for $\phi(L)$. For $m = 1$, the equation shows that $V(1)$ is optimal against all one-sided AR(1) alternatives: varying $q$ in this case traces out all AR(1) spectra. Equation (7) also reveals the fact that the power of the test based on the lower tail of $V(m)$ is concentrated against mean reverting alternatives. Note that for $m$ of any horizon, the spectra of $V(m)$-best class models with $q > 0$ have global minimum at frequency zero. This is a sufficient condition for mean reversion under most reasonable definitions, implying that each $V(m)$ test is an optimal test for some type of mean reversion.

Some general lessons about the $V(m)$-best class also become apparent from (7). For example, the variety of models the test is best against at horizon 3 is shown in Figure 1, and Figure 2 illustrates how the class changes as the horizon grows. The spectra show damped fluctuations as the frequency increases; lower spectral power at frequency zero and higher values of $m$ lead to a flatter spectrum at high frequencies.[6] These figures reveal a formal basis for the argument that large $m$ is to be preferred to small when testing for mean reversion: if the goal is to detect a trough in the spectrum near zero in cases where little or no high-frequency dependence is expected, then a larger $m$ is preferred to smaller (ignoring sample size considerations).

An important question in implementing the $V(m)$ test is how to choose $m$ to maximize power against particular alternatives. Considering the FVR class as a whole, one would want to know how to choose both the coefficients and order of $\phi(L)$ to maximize power. Since choosing $m$ for the $V(m)$ test involves the same considerations as choosing a general $\phi(L)$, however, I focus on this simpler issue. The basic idea is to choose the $m$ for which the alternatives of interest lie closest in some metric to a $V(m)$-best class.

In some cases, such as the AR(1) case below, simply plotting the spectra of the alternative model of interest against the spectra of best-class models will often suggest a reasonable $m$. For a more formal approach, the best criterion is

[6] The AR parameters of the $V(m)$-best class models are always of the same sign as $q$ (positive for models in Figures 1 and 2) and monotonically declining in absolute value.

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

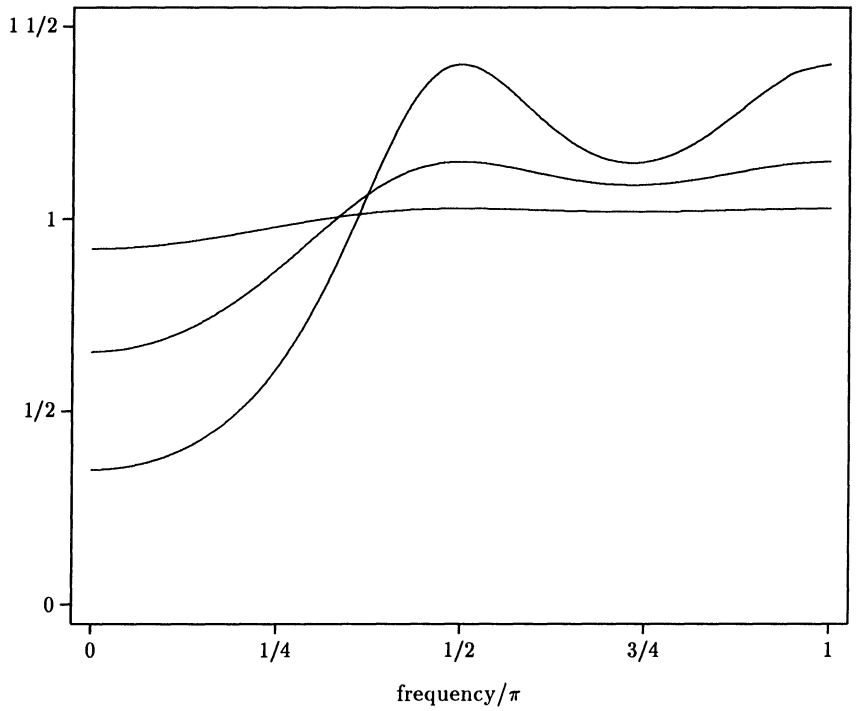


FIGURE 1.—Spectral densities of three members of the $V(3)$-best class. From greatest to least density at zero, these are generated by $q = 0.009, 0.006, 0.240$.

probably directly to compute the power of $V(m)$ tests of various horizons against the alternatives of interest at the relevant sample and test sizes. For stationary ARMA alternatives, this can be done using the exact distribution discussed above: simply compute the exact power for some range of $m$ and choose the one that maximizes power.[7]

A simpler approach is to select $m$ based on some asymptotic criterion. For example, Richardson and Smith (1989) have suggested that $m$ be chosen to maximize the approximate slope of the test against the alternative of interest (see also Jegadeesh (1991)). This approach has two weaknesses. First, it ignores sample size. Because raising $m$ effectively uses up observations as initial conditions, the optimal $m$ in large samples need not be optimal in smaller ones. This difference between the optimal small-sample and large-sample $m$ can be quite important when, as in the business cycle literature, researchers want to test for mean reversion at horizons that are a nontrivial fraction of sample size. The second weakness of the approach is that, as Bahadur (1967) emphasized, the approximate slope in many cases does not pick the test that has maximum exact slope.

[7]Given the ease of computing the exact power, even choosing a general $\phi(L)$ becomes a straightforward nonlinear optimization exercise of choosing the $\phi_i$ to maximize power, given the alternative and sample and test sizes.

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

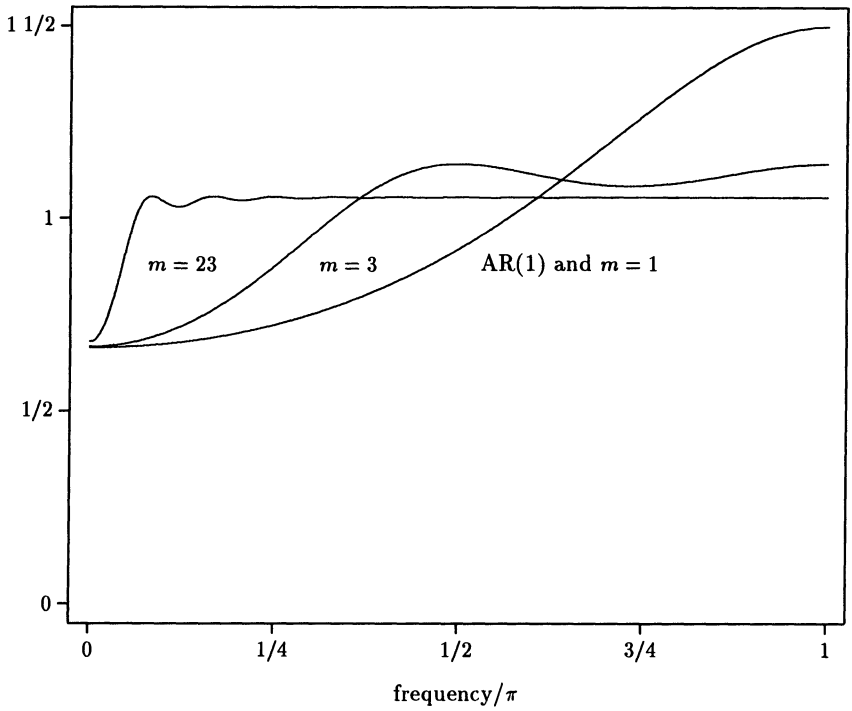


FIGURE 2.—Spectral densities of an AR(1) model and a member of each $V(m)$-best class, $m = 1, 3, 23$. The AR parameter is .2. From smallest to largest $m$ these are generated by $q = 0.2, 0.057, 0.007$.

While taking care in selecting $m$ may be cumbersome, a simple example illustrates that picking $m$ so that the alternative of interest is near the $V(m)$-best class can pay substantial rewards. Consider an AR(1) alternative, which lies in the $V(1)$-best class, and falls increasingly far from the $V(m)$-best class as $m$ increases (see Figure 2). The Monte Carlo evidence of Lo and MacKinlay (1989) illustrates that raising $m$ from its optimal value 1, even by amounts that are small relative to sample size, can be quite costly. For example, at sample size 256, raising $m$ from 1 to 7 lowered power from 89 to 50 percent.[8] In this case, however, choosing $m$ by simple visual comparison of the spectra of the $V(m)$-best class and alternative, as in Figure 2, would easily allow one to choose $m = 1$.

A final question involves the relation between FVR tests and tests for unspecified AR($m$) dependence, such as likelihood ratio, Box-Pierce $Q$, or Godfrey's LM tests. Analytically, the null for the general tests is defined by $m$ zero restrictions on the AR parameters of the general AR($m$) model. In contrast, the null for the FVR test is specified by a single restriction ($q = 0$) on

[8] See Lo and MacKinlay (1989, Table 5e). The AR parameter was $-0.2$, and the test size was 5 percent.

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

the more narrow class of AR($m$) alternatives. Thus, relative to the general tests, the FVR test gains $m-1$ degrees of freedom, but sacrifices power against AR($m$) alternatives far from the best class. The Monte Carlo work of Lo and MacKinlay also provides some evidence on this point. For the five alternative models investigated, the $V(m)$ test dominated or equalled the power of the Box-Pierce $Q(m)$ test for the vast majority of sample sizes and choices of $m$.

Overall, Propositions 1 and 2 provide a great deal of information regarding the variance ratio test for mean reversion. For practitioners, the results reveal both the strengths and weaknesses of the test. They verify that no more powerful test can be found for the forms of mean reversion in the best class. On the other hand, in revealing where the test's power is focused, the results also reveal what forms of mean reversion the test may have missed.

## 4. CONCLUSION

This paper presents a class of filter variance ratio statistics and a simple relation between the filter upon which a test is based and the models against which it will have optimal properties. This result makes trivial the analysis of existing FVR tests, a fact illustrated above in deriving the optimality properties of the variance ratio test for mean reversion.

The FVR framework also provides a straightforward approach to creating tests for specified classes of dependence. Such specification tests have several attractive features. Pragmatically, the simple form of the statistics makes them easy to calculate, and $p$-values can be calculated by evaluating the exact distribution of the statistic. Because the tests have power concentrated against dependence at particular frequencies, they may have considerably greater power against forms of dependence of interest than do more general specification tests.

*International Finance Division, Board of Governors of the Federal Reserve System, Washington, D.C. 20551, U.S.A.*

*Manuscript received December, 1989; final revision received March, 1992.*

## APPENDIX

LEMMA 1: (i) *Assume $\sigma^2\Sigma$ is the $(T\times T)$ covariance matrix of an AR($m$) model, with $T>2m$. Except for the $(m\times m)$ blocks in the northwest and southeast corners, $\Sigma^{-1}$ is symmetric and Toeplitz with zero on the jth diagonal for $j>m$. If the AR parameters of the process are given by $\rho_j$, $j=0,\dots,m$, $(\rho_0=1)$ then the element on the jth diagonal of $\Sigma^{-1}$, $j\leqslant m$, is given by $g_j$ in*

$$\begin{bmatrix} g_0 \\ \vdots \\ \vdots \\ \vdots \\ g_m \end{bmatrix} = \begin{bmatrix} \rho_0 & \cdots & \cdots & \rho_m \\ \vdots & & \cdot & 0 \\ \vdots & \cdot & \cdot & \vdots \\ \rho_m & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \rho_0 \\ \vdots \\ \vdots \\ \vdots \\ \rho_m \end{bmatrix}. \tag{8}$$

(ii) *Define the vector $g^*$ as the left-hand side of* (8) *when $\rho_j=\phi_j$ for some filter $\phi(L)$. With $T>2m$, except for $(m\times m)$ blocks in the northwest and southeast corners, $\alpha(I+q\Phi'\Phi)$ is symmetric*

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

*and Toeplitz with zero on the jth diagonal for $j > m$. The jth diagonal of $\alpha(I + q\Phi'\Phi)$ for $j \leqslant m$ is given by $\tilde{g}_0 = \alpha(1 + qg_0^*)$ and $\tilde{g}_j = \alpha q g_j^*$, $j = 1, \ldots, m$.*

(iii) *If $T > 2m$ and $\sigma^2\Sigma_q$ is a $(T \times T)$ covariance matrix from a process solving* (3), *then $\alpha(I + q\Phi'\Phi) - \Sigma_q^{-1} = 0^*$ for some $\alpha > 0$, where $0^*$ is a matrix of zeros except for the $(m \times m)$ blocks in the northwest and southeast corners.*

PROOF: The first two parts simply describe the structure of matrices; they can be verified directly and are described more extensively in Faust (1988). Using (i) and (ii), (iii) is true whenever the values on the diagonals of the Toeplitz portions of the matrices are equal, e.g., when $\tilde{g} = g$. Substituting the definitions of $\tilde{g}$ and $g$ reveals that this equation is a time domain statement of (3), which holds by assumption.

PROOF OF PROPOSITION 1: Throughout, consider (3) with $z = e^{-i\theta}$. First show necessity of the lower bound on $q$. Under a solution for $\rho_q(L)$ that is a stationary AR process, the left-hand side of (3) is the spectrum of an MA($m$) process and, hence, positive. For finite $m$, $\phi(z)\phi(z^{-1})$ is nonnegative, implying the right-hand side is positive at all frequencies if and only if the bound on $q$ is satisfied. This necessary condition is also sufficient for the right-hand side to define the polynomial of a stationary AR process, since under this condition the right-hand side of (3) is strictly positive, and, hence, is the spectrum of an MA($m$) process with no unit roots. This process has a unique invertible representation, which is the solution for $\rho_q(L)$ that is sought. (Notice further, that beginning with an arbitrary $\rho_q(L)$ for a stationary process, the same logic shows that there exists a $\phi(L)$ that solves (3).)

PROOF OF PROPOSITION 2: Part (i). Given the ergodicity of $R$ under $H_N$ and $H_K$, the following results can be derived using a strong law of large numbers such as White's Theorem 3.34 (1984). Under $H_K$ the maximum likelihood estimate of the mean of $R$ is $\tilde{R} \equiv (i'\Sigma^{-1}i)i'\Sigma^{-1}R$. Now $\alpha(1 + q\nu_\phi)$ can be written

$$\frac{\left(R - \tilde{R}i + (\tilde{R} - \bar{R})i\right)'(\Sigma^{-1} + 0^*)\left(R - \tilde{R}i + (\tilde{R} - \bar{R})i\right)}{(R - \bar{R}i)'(R - \bar{R}i)}$$

with $0^*$ as in Lemma 1. Subtracting $LR$ and multiplying by $T^{1/2}$ yields

$$\frac{T^{1/2}(\tilde{R} - \bar{R})^2\left[i'\Sigma^{-1}i/T\right] + 2T^{1/2}(\tilde{R} - \bar{R})\left[(R - \tilde{R}i)'\Sigma^{-1}i/T)\right]}{\left[(R - \bar{R}i)'(R - \bar{R}i)/T\right]} \tag{9}$$

plus terms in $0^*$ which go to zero. Each term in square brackets converges almost surely, and the limit of the denominator is strictly positive. Consider $T^{1/2}(\tilde{R} - \bar{R})$. All but the first and last $m$ rows of $\Sigma^{-1}$ have the same row sum, implying this term can be written

$$T^{-3/2}w_0 \sum_{t=m+1}^{T-m} R_t + T^{-1/2} \sum_{j=1}^{m} w_{Tj}\left[R_j + R_{T+1-j}\right]$$

all divided by $i'\Sigma^{-1}i/T$, for some finite $w_0$ and nonstochastic weights, $w_{Tj}$, which converge for each $j$. This entire term goes almost surely to zero, implying $T^{1/2}(\tilde{R} - \bar{R})^2$ does as well, and completing the proof.

Part (ii). Because Crowder's (1976) conditions are met, the sequence of tests $\{LR(q^*)\}$ is locally most powerful invariant under the assumptions here (Engle (1984)). The result follows from this fact and part (i).

Part (iii). Suppose the null is $\theta \in \Theta_N$, the alternative is $\theta \in \Theta - \Theta_N$, and a statistic, $s_T$, is such that large values are considered significant in rejecting $H_N$. Define $F_\Theta^T(\gamma) = \inf_{\theta \in \Theta} \Pr_\theta(s_T \leqslant \gamma)$. If it exists, the exact slope of the test for any parameter vector in the alternative is plim $-2/T(\ln[1 - F_{\Theta_N}^T(s_T)])$, when $s_T$ is generated under the alternative. The fact that $LR$ has the maximum possible exact slope equal to $-\text{plim}\ln(LR)$ can be shown following the proof of Bahadur (1967) with slight modifications for dependence. Thus, it is sufficient to show that the slope of $\nu_\phi$ is at least as great as the slope of $LR$.

Define $\tilde{\nu}_\phi \equiv \alpha(1 + q\nu_\phi)$ where $\alpha$ and $q$ are (as in part i) such that plim $\tilde{\nu}_\phi$ = plim $LR$, and choose the statistic $s_T = -\ln(\tilde{\nu}_\phi)$, which is equivalent to $\nu_\phi$. Select a particular $\theta_n \in \Theta_N$ and $\theta_k \in \Theta - \Theta_N$. By Markov's inequality,

$$\Pr_{\theta_n}(s_T > \gamma) = \Pr_{\theta_n}\left(\tilde{\nu}_\phi^{-T/2} > e^{\gamma T/2}\right) \leqslant e^{-\gamma T/2}E_{\theta_n}\left[\tilde{\nu}_\phi^{-T/2}\right].$$

This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms

For sufficiently large $T$, the final expectation is positive and less than one. Thus, $-2/T(\ln[1-F_{\theta_n}^T(\gamma)]) \geqslant \gamma$. Remembering that $\theta_n$ was arbitrary, substituting $\gamma = s_T$ generated under $H_k$, and taking the limit gives the required result for $s_T$ and, hence, $\nu_\phi$.

## REFERENCES


ANDERSON, T. W. (1948): "On the Theory of Testing Serial Correlation," *Skand. Aktuar Tidskr.*, 31, 88–116.

BAHADUR, R. R. (1967): "Rates of Convergence of Estimates and Test Statistics," *Annals of Mathematical Statistics*, 38, 303–324.

COCHRANE, JOHN H. (1988): "How Big is the Random Walk in GNP?" *Journal of Political Economy*, 96, 893–920.

COGLEY, TIMOTHY (1990): "International Evidence on the Size of the Random Walk in Output," *Journal of Political Economy*, 98, 501–518.

CROWDER, M. J. (1976): "Maximum Likelihood Estimation for Dependent Observations," *Journal of the Royal Statistical Society*, Series B, 45–53.

DAVIDSON, RUSSELL, AND JAMES G. MACKINNON (1987): "Implicit Alternatives and the Local Power of Test Statistics," *Econometrica*, 55, 1305–1329.

DURBIN, J., AND G. S. WATSON (1950): "Testing for Serial Correlation in Least Squares Regression, I," *Biometrika*, 37, 409–428.

——— (1971): "Testing for Serial Correlation in Least Squares Regression, III," *Biometrica*, 58, 1–19.

ENGLE, R. F. (1984): "Wald, Likelihood Ratio, and Lagrange Multiplier Tests," in *Handbook of Econometrics*, Vol. 2, ed. by Zvi Griliches and Michael D. Intriligator. New York: North Holland, 776–826.

FAUST, JON (1988): "The Variance Ratio Test: Statistical Properties and Implementation," Research Working Paper, Federal Reserve Bank of Kansas City, 88–08.

GEWEKE, JOHN (1981): "The Approximate Slopes of Econometric Tests," *Econometrica*, 49, 1427–1442.

IMHOF, P. J. (1961): "Computing the Distribution of Quadratic Forms in Normal Variables," *Biometrika*, 48, 419–426.

JEGADEESH, N. (1991): "Seasonality in Stock Price Mean Reversion: Evidence from the U.S. and the U.K.," *Journal of Finance*, 36, 1427–1444.

KIM, M., C. NELSON, AND R. STARTZ (1988): "Mean Reversion in Stock Prices? A Reappraisal of Empirical Evidence," Manuscript, University of Washington—Seattle.

LO, ANDREW W., AND A. CRAIG MACKINLAY (1988): "Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test," *Review of Financial Studies*, 1, 41–66.

——— (1989): "The Size and Power of the Variance Ratio Test in Finite Samples, A Monte Carlo Investigation," *Journal of Econometrics*, 40, 203–238.

PAN, JIE-JIAN (1968): "Distribution of the Noncircular Serial Correlation Coefficients," *Am. Math. Soc. and Inst. Math. Statist. Selected Translations in Probability and Statistics*, 7, 281–291.

POTERBA, JAMES M., AND LAWRENCE H. SUMMERS (1988): "Mean Reversion in Stock Returns: Evidence and Implications," *Journal of Financial Economics*, 1, 27–60.

RICHARDSON, M. (1988): "Temporary Components of Stock Prices: A Skeptic's View," Manuscript, Wharton School of Business.

RICHARDSON, M., AND T. SMITH (1989): "Robust Power Calculations with Tests for Serial Correlation in Stock Returns," Manuscript, Wharton School of Business.

——— (1989): "Tests of Financial Models in the Presence of Overlapping Observations," Working Paper, Futures and Options Research Center, Duke University, 89–106.

RICHARDSON, M., AND J. STOCK (1989): "Drawing Inferences From Statistics Based on Multiyear Asset Returns," *Journal of Financial Economics*, 1, 323–348.

WATSON, MARK W. (1986): "Univariate Detrending Methods with Stochastic Trends," *Journal of Monetary Economics*, 18, 49–75.

WHITE, HALBERT (1984): *Asymptotic Theory for Econometricians*. New York: Academic Press.


This content downloaded from
124.217.188.109 on Sat, 07 Feb 2026 06:28:55 UTC
All use subject to https://about.jstor.org/terms